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Title: Clinical Pharmacy-Standard Of Procedure 3.3-Management of adverse drug reaction

Clinical Pharmacy Standard of Practice

3.3.Adverse Drug Reaction Management

OBJECTIVE AND DEFINITION

- **ADR management** is the prevention, detection, assessment, management, documentation and reporting of ADRs.
 - ADR is a response to a medicine that is noxious and unintended
 - which occurs at doses normally used or tested in humans for the prophylaxis, diagnosis or therapy of disease, or for the modification of physiological function.
 - An unexpected therapeutic response may be a side effect but not an adverse reaction.

POLICY AND PROCEDURE

- ADR management involves the detection, assessment and correlation, management, documentation and prevention of ADRs.
 - The emphasis of ADR management is on preventing ADRs and preventing re-exposure of patients who have already experienced an ADR.
- **Detecting ADRs**
 - Identify and monitor patients susceptible to ADRs. Particular patient groups include:
 - those who have previously experienced ADRs
 - those with multiple disease processes
 - those on a large number of medicines

- those treated with medicines known to have a high incidence of adverse effects
 - those treated with medicines known to be associated with serious adverse effects
 - those treated with drugs with a low therapeutic index
 - those taking medicines with the potential for multiple interactions
 - those with abnormal investigation results.
- Orders for single doses of medicines, such as antihistamines, adrenaline and corticosteroids may indicate that an adverse reaction has occurred.
- Encourage nursing and medical staff and patients/ carers to report any suspected ADRs.
- **Suspected Adverse Drug Reactions**
 - When suspected, assess the details of the ADR in the context of patient-specific and medication-related factors. Patient-specific factors include:
 - age, gender, race, organ function, height, weight
 - diagnosis and other relevant comorbidities prior to reaction
 - previous exposure to suspected or related drug(s). Medication-related factors include:
 - non-prescription drugs, complementary and alternative therapies
 - recently ceased medicines
 - suspected causative drug (name, dose, route of administration, manufacturer, batch, date and time commenced, date and time discontinued [if applicable], indication).
- Comprehensive adverse reaction details include:
 - description of the reaction
 - time of onset and duration of reaction
 - complications and sequelae
 - treatment and outcome of treatment
 - relevant investigation results or autopsy report.
- When assessing the likelihood that a suspected ADR was caused by a particular drug, review the relevant literature and where appropriate consult with other health professionals.
- Causality of a suspected medicine with an adverse reaction may be:
 - **Certain** – a clear association is established between administration of the drug and the reaction,
 - the results of investigations confirm that there is a relationship between administration of the drug and the reaction,
 - the reaction recurs on re-exposure to the drug, or
 - the reaction is known to occur with the suspected drug.
 - **Probable** – the reaction is known to occur with the suspected drug, and there is a possible association between the reaction and administration of the drug,
 - the reaction resolves or improves on withdrawal of the suspected drug and other medicine therapy remains unchanged, or
 - an uncommon clinical event occurs in the absence of other potentially causative factors.
 - **Possible** – an alternative explanation for the reaction exists, more than one drug is suspected,
 - recovery follows withdrawal of more than one medicine or
 - the association between the reaction and administration of the medicine is unclear.
 - **Doubtful** – another cause is more likely to have accounted for the clinical event, e.g. underlying disease.
- **Managing and Reporting Adverse Drug Reactions**
 - The likelihood of the suspected medicine(s) having caused the reaction and the clinical significance of the reaction are considered when assessing whether to continue treatment with the suspected medicine(s).
 - In many cases a reasonable alternative treatment will be available.
 - Recommend treatment options for the ADR and, if appropriate, recommend alternative treatments.
 - Important issues to consider when managing ADRs include the:
 - patient's condition
 - requirement for therapy (whether treatment can be ceased)

- reaction
 - relative efficacy and safety of other therapeutic alternatives
 - prophylactic use of other medicines to prevent future adverse reactions
 - consideration of rechallenge and desensitisation for the medicine.
- Ensure all suspected ADRs are communicated appropriately by:
 - documenting in the health record, including electronic prescribing or dispensing system,
 - if appropriate attach relevant alert notices/stickers to medicine administration records and health record.
 - documenting in the MMP or equivalent
 - notifying medical and nursing staff (including the original prescriber)
 - supplying a record of the ADR to patients/carers when potential for re-occurrence is deemed significant
 - documenting ADRs according to local policy.
- Likelihood of future ADRs can be minimised by:
 - documenting ADRs to avoid patient re-exposure on all medicine orders, including noting when there are no known ADRs
 - documenting ADRs in the MMP and the electronic health record
 - monitoring patients at risk of ADRs
 - careful use of medicines that have a high incidence of or are known to cause serious adverse effects,
 - ensuring patient is given a hospital alert wrist band where appropriate,
 - providing education and information, including alert cards, to patients who have experienced serious ADRs.

Reference:

Adapted with modification from:

George Taylor, et. al. Standards of Practice for Clinical Pharmacy Services. Journal of Pharmacy Pharm Pract Res 43, No. 2 (suppl), 2013.

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Developed By: Zaher Al Salmi

Reviewed By: Ph. Asma Alhindi

Approved By: Faisal Al Harthi

Written By: Zaher AlSalmi

Checked By: Asma Alhindi

Authorized by: FAISAL AL-HARTHI



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